What a reward model can show depends on who stopped the episode: failure judges and monitors on a low-cost arm

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Abstract

Reward models that score robot rollouts are also used as failure detectors, and the baseline such a detector must beat depends on who stopped the episode. On 2,047 public SO-101 rollouts from 7 policies, where runs end at a time limit, plain episode length separates failure from success at AUROC 0.942 within policy. It does not need the timeout: watched only up to half of the longest episode, length still gives 0.851, because failed runs are slow rather than cut off. Both zero-shot reward models we test score below that baseline, the better one at 0.821 within policy. Our own rollouts, which an operator stopped by judgment, reverse the picture. Length falls to 0.56, neither reward model beats the policy-identity null, and what works is a judge at rest. A robot-trained 27B judge passes its pre-registered bar there, though at an operating threshold chosen in-sample, while every mid-episode monitor we tested misses. We recommend reporting a within-policy length baseline and naming the dataset's stopping rule.

1 Introduction

A reward model that scores robot rollouts is also a failure detector, and papers evaluate it as one. Such an evaluation has a hidden variable: the dataset's *stopping rule*, meaning who decides that an episode is over.

When a benchmark stops failed runs at a time limit, the length of an episode encodes its outcome. A detector should then be read against two baselines that the stopping rule hands out for free: elapsed length, and the identity of the policy that ran, which we call the policy-identity null. When an operator stops each run by judgment, both baselines collapse and the detection task changes with them.

None of the reward-model papers in our library reports how far plain episode length gets on its own data. We measured it on a public SO-101 benchmark [12] and on our own rollouts, with the length analyses frozen before any number existed. Against it we set two zero-shot reward models [5, 1] and every judge and monitor we had run on our 88 operator-stopped rollouts, with each threshold chosen in-sample marked as such.

2 Data and labels

ArmnetBench v0.1 [12] covers the SO-101 with 8 tasks and 7 policies. We use its 2,047 policy rollouts with a success or failure class (515 successes, 1,532 failures) and leave out the teleoperated episodes. One task, pushing a Displayport cable (270 rollouts), has rollouts of a single class and so no per-task AUROC. Our own data are 88 labelled bowls-task episodes on one SO-101 arm with a Jetson Orin NX, and the 50-episode subset (42 successes, 8 failures) that every monitor gate used. The operator labelled them and was not blind to the policy.

3 The length baseline

On ArmnetBench, episode length is a strong failure detector, and not only because it tells the policies apart (Table 1, Fig. 1). Pooled over policies it reaches 0.938. Within each policy it ranges from 0.91 to 0.96, or 0.942 pair-weighted, with a CI of 0.927–0.956.

Both reward models fall below it. Robometer-4B [5], run zero-shot as its authors ship it, scores 0.818 pooled and 0.821 within policy. TOPReward [1], with Qwen3-VL-8B [11] and no training, scores 0.661 within policy; its paired difference to length has a 95 % CI of -0.31 to -0.25. The protocols fixed each model's primary score before any score existed: Robometer's success probability and the success score of TOPReward's paper. Read descriptively from the same runs, Robometer's last-progress head does better at 0.838 pooled, and TOPReward's full-video Yes/No component reaches 0.678 within policy. Neither reaches length.

Robometer's within-policy number has no episode-bootstrap CI. Its per-episode scores stayed on an older

*Code, result files and the checks that regenerate every number: `github.com/ravediamond/paper-reward-models-stopping-rule`. The frozen protocols are copied verbatim under `repro/`; the bar commits named in the paper belong to the authors' research repository `ravediamond/road-to-robotics`.

signal	pooled	within policy
episode length	0.938 (0.925–0.951)	0.942 (0.927–0.956)
length seen up to 50 % of the longest run	—	0.851 (0.829–0.873)
Robometer-4B, success probability	0.818 (0.790–0.843)	0.821 (per-policy 0.79–0.87)
Robometer-4B, last progress (descriptive)	0.838 (no CI)	—
TOPReward, success score	0.653 (0.623–0.684)	0.661 (0.631–0.692)
TOPReward, Yes/No component (descriptive)	—	0.678

Table 1: ArmnetBench, failure AUROC with 95 % episode-bootstrap CIs. The Robometer within-policy entry is pair-weighted over its per-policy block and carries the per-policy range instead of a CI (per-episode scores not stored); the last-progress entry is descriptive (the protocol's primary score was the success probability).

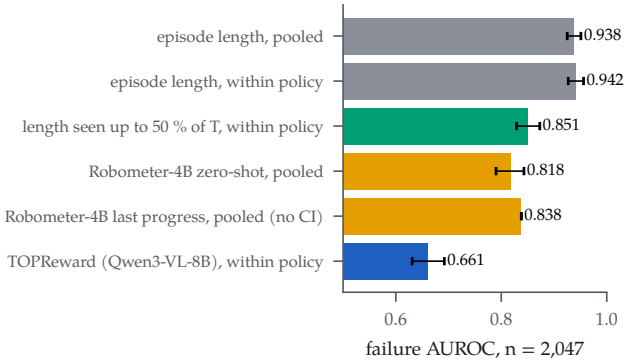


Figure 1: Failure AUROC on ArmnetBench (95 % CIs where available): episode length against two zero-shot reward models (Robometer-4B, TOPReward with Qwen3-VL-8B). The last-progress bar is descriptive and carries no CI. The Displayport-cable task contributes rollouts of one class only and has no per-task AUROC.

machine and were not copied with the run, so we give the per-policy range (0.79–0.87) instead, and Robometer is missing from the truncation analysis of §4.

4 Is it the timeout?

If length only reflected the timeout, its signal would sit in the few runs that hit it. We take each task's longest episode as its timeout, and only 8.7 % of failures run that long. At half that length, 88 % of failures are still running but only 19 % of successes. Length observed up to that cut gives 0.851 within policy (CI 0.829–0.873; Fig. 2), and TOPReward on frames up to the same cut reaches 0.667. Failed rollouts are slow, not timed out. Both readings the protocol fixed in advance hold: length is a real within-policy null, and it is more than the timeout.

On our own operator-stopped rollouts the same baseline is at chance: 0.56 on the 69 rollouts TOPReward scored and 0.52 on all 88. The baseline that matters here

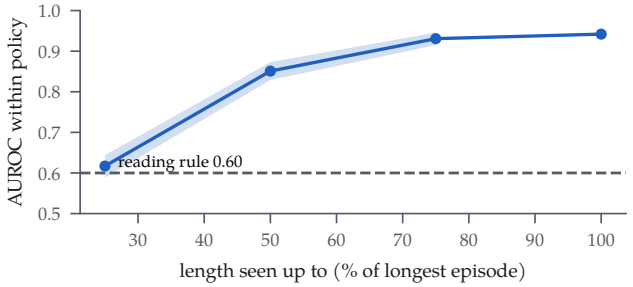


Figure 2: Length observed up to a fraction of the task's longest episode, AUROC within policy (band: 95 % CI).

is the policy-identity null, because 17 of the 25 failures in the 69 come from one policy, TurboVLA [16]. Neither reward model beats it. TOPReward scores 0.82 (CI 0.70–0.93) against 0.87 for policy identity alone, and 0.78 on the 50 episodes without TurboVLA. Robometer scores 0.577 against 0.878 on all 88. Which baseline a detector must beat depends on how the episodes ended, so it has to be reported per dataset.

5 Judges and monitors on our rollouts

Table 2 reports every judge and monitor the same way: a threshold-free AUROC where one is defined, then failures caught and successes flagged at the operating point. The judges at rest have their operating thresholds chosen in-sample on the same 50 episodes. Their AUROCs are not.

A judge that sees the finished episode works on these rollouts, and the input format matters more than the robot fine-tune. A deterministic top-camera check (colour masks and geometry, no learning) is right on 94.3 % of judged episodes, with 0 false positives (Wilson 0–13 64-frame two-view clip and NVIDIA's prompt, Flora-Rewarder-27B [8] reaches a threshold-free AUROC of 1.00 (bootstrap CI 1.00–1.00). At its in-sample operating point it flags 8 of 8 failures at 2 of 42 successes (Wilson on the false alarms 1–16 bar of at least 6 of 8 failures at no more than 2 of 42 successes. Its base model Qwen3.5-27B [10], without the robot fine-tune, passes the same bar on the same input: AUROC 0.91 (0.78–1.00), 7 of 8 at 2 of 42. The smaller Qwen3.5-9B [10] and Gemini 3.8 Flash [2] fail it, the 9B by flagging 16 successes and Gemini by one false alarm too many.

No mid-episode monitor meets its bar of at least 5 of 8 failures at no more than 1 of 20 false interventions (Fig. 3). The surprise of a separate world model, trained in the LeWorldModel recipe [7], comes closest with 2 of 8 at 2 of 42 (AUROC 0.86). A Gemini sliding-window judge catches 0. The action-conditioned risk head on the world model's latent has a history. Its first version reached AUROC 0.93, but it read elapsed time, so we withdrew that result and pre-registered a second version without time

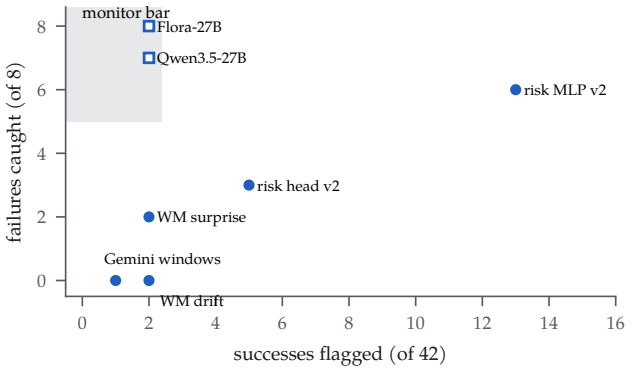


Figure 3: Fifty own rollouts: failures caught (of 8) against successes flagged (of 42). Circles: mid-episode monitors; shaded: their bar ($\geq 5/8$ at $\leq 1/20$). Squares: judges at rest (one score per finished episode) at their in-sample operating point; the monitor bar does not apply to them.

or length. The second version catches 3 of 8 at 5 of 42, with AUROC 0.79 against 0.71 when its actions are shuffled, and its MLP variant catches 6 of 8 but flags 13 of 42.

6 Related work

Robometer [5] and TOPReward [1] are the two reward models we test. Robometer scales a general-purpose reward model through trajectory comparisons, and TOPReward reads the token probabilities of a video-language model as hidden rewards. A survey of verifiers for robot policies [14] maps the wider family. Mid-episode failure detection is its own line of work, and our monitors borrow from it. FAIL-Detect [17] and SAFE [3] learn uncertainty- and feature-based failure scores for imitation and VLA policies. FARM [9] reads failure signals from the internal states of a frozen world model, close to our world-model monitors, and ReAP [6] predicts action-conditioned risk, the idea behind our risk head. VLA-Corrector [13] verifies stages to drive prompt-based recovery, Spotter [4] pairs the policy with a reflecting VLM, and REMORA [15] tracks task progress through errors and recovery with a recurrent distributional model. ArmnetBench [12] and a standardized SO-101 VLA benchmark [18] evaluate policies on the same low-cost arm. None of these works reports the elapsed-length baseline under the dataset's stopping rule, and that comparison is what this paper adds.

7 Limitations

The results cover two task families, ArmnetBench's eight tasks and our bowls task. Our labels are the operator's and not blind to the policy. The risk head is trained and tested on the same 50 episodes: five folds by episode keep episodes from leaking, but tasks and policies do leak. Robometer's within-policy and truncation analyses lack an episode-bootstrap CI because its per-episode scores were not stored with the run. TOPReward's paper does not give its prefix count, frames per query or Yes/No wording, so those are our choices. The operating points of Flora-Rewarder-27B, Qwen3.5-27B and Gemini 3.8 Flash are chosen in-sample on the 50 episodes they are scored on, though their AUROCs are not, and fresh episodes scored at those frozen thresholds would test them. 17 of the 25 failures in our 69 rollouts are TurboVLA's, so on our set any score that reads the policy gets the policy-identity null for free. The timeout is approximated by each task's longest episode.

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method	kind	AUROC	caught / 8	flagged / 42	threshold
colour judge (no learning)	at rest	— (94.3 % right, 0 FP, 5 FN)	—	—	fixed rule
Flora-Rewarder-27B	at rest	1.00 (1.00–1.00)	8	2	in-sample
Qwen3.5-27B, same input	at rest	0.91 (0.78–1.00)	7	2	in-sample
Qwen3.5-9B, same input	at rest	0.75	7	16	in-sample
Gemini 3.8 Flash, same input	at rest	0.96	8	3	in-sample
Gemini sliding windows	monitor	—	0	1	fixed before scoring
world-model surprise	monitor	0.86	2	2	fixed before scoring
world-model latent drift	monitor	0.30	0	2	fixed before scoring
risk head, logistic (v2)	monitor	0.79	3	5	5-fold by episode
risk head, MLP (v2)	monitor	—	6	13	5-fold by episode

Table 2: Judges and monitors on 50 own operator-stopped rollouts (42 successes, 8 failures). Judges score the finished episode; monitors score mid-episode windows against the monitor bar ($\geq 5/8$ caught at $\leq 1/20$ false interventions). "In-sample" marks an operating threshold chosen on these same 50 episodes.

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analysis	repro folder	expected files	protocol
length baseline, truncation	repro/length-null	length_armnetbench.json, counts_ci.json	LENGTH_PROTOCOL.md, frozen `1a69f30`
Robometer on ArmnetBench	repro/robometer-judge	g9a_ab_complete.json	PROTOCOL.md (primary score fixed there)
Robometer and length on own rollouts	repro/robometer-judge	g9a_own.json	PROTOCOL.md
colour judge	repro/robometer-judge	g9c3_test_results.json	PROTOCOL.md
TOPReward, both datasets	repro/topreward	topreward.json, own_rollout_labels.csv	TOPREWARD_PROTOCOL.md, `6241570`
judges at rest, input-format gate	repro/flora-input-format	g_flora_summary.json, g_flora_abl_summary.json	PROTOCOL.md
window judge, world-model monitors	repro/six-monitors	judge_tau.json, w1a_summary.json	PROTOCOL.md
action-conditioned risk head	repro/reap	monitors.json, monitors_v2.json	PROTOCOL.md

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Appendix: from plain name to analysis

Every number is recomputable from the repository; this table maps each analysis to its files.